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Capstone Project Report
on
PLC-Based Automatic Conveyor Sorting System

Submitted in partial fulfillment of the requirements
*of the **Third Year** of*
Bachelor of Technology
In
Electronics and Computer Science

by

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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
UNIVERSITY OF MUMBAI
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This is to certify that, following students of T. Tech in Electronics and Computer Science Department. They have satisfactorily completed the requirements for the courses Programmable Logic Controllers and SCADA prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai),** while working on "PLC-Based Automatic Conveyor Sorting System"

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Abstract

This project presents a PLC-Based Automatic Conveyor Control System designed to demonstrate the automation of object movement, object counting, and conveyor operation using Programmable Logic Controller (PLC) technology. The proposed system uses a Mitsubishi FX5U PLC programmed through MELSOFT GX Works3 using Ladder Diagram logic. In conventional material-handling systems, manual monitoring and counting can be time-consuming and may lead to counting errors. An automated control system can improve operational consistency by controlling conveyor operation through programmed logic and registering object-detection signals. The PLC receives input signals from start, stop, reset, and object-detection devices and processes them according to the programmed sequence. The resulting output signals control the conveyor motor interface and provide status indications. The proposed ladder logic includes start/stop control, a master-run relay, object-counting logic, counter-reset functionality, and a count-completion indication. A reset object count can be used to demonstrate automatic stopping after the required number of detection pulses. For the available PLC training setup, pushbuttons and output indicator lamps may be used to demonstrate the logic in place of an actual object sensor and conveyor motor, subject to the available wiring. The project provides a practical introduction to industrial automation, digital input/output handling, internal relays, counters, and ladder programming. It also provides a foundation for future implementation using a physical conveyor, object sensor, motor driver, and suitable safety devices.

Keywords: PLC, Mitsubishi FX5U, GX Works3, Ladder Diagram, Conveyor Automation, Object Detection, Object Counting, Industrial Automation.

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CHAPTER 1: OVERVIEW

1.1 Introduction

Industrial automation improves productivity and consistency in manufacturing and material handling operations. Conveyor systems transfer objects between workstations in manufacturing, packaging, warehousing, and assembly lines. This project proposes a conveyor-control sequence using the Mitsubishi FX5U PLC and MELSOFT GX Works3. A start command initiates operation, a stop command stops the run sequence, and valid object-detection pulses increment a counter. At a preset count, the program can inhibit running and activate a completion indication. The design demonstrates contacts, coils, internal relays, counters, reset logic, and PLC monitoring.

1.2 Background

Manual conveyor operation and counting may require continuous attention and can result in missed or repeated counts. A PLC provides flexible sequence control by reading inputs, executing a program, and updating outputs. In a full system, a compatible sensor produces a signal for each passing object. Edge detection or pulse conditioning is important to prevent repeated counts while a sensor remains active.

1.3 Importance of the Project

- Automates start/stop control through programmed logic.
- Counts valid object-detection events using a PLC counter.
- Reduces reliance on manual counting.
- Provides a count-completion indication at a preset value.
- Demonstrates industrial PLC programming and monitoring.
- Can be extended with sensors, HMI, and sorting mechanisms.

1.4 Perspective of Stakeholders and Customers

- Manufacturing industries can use conveyor automation for consistent movement and counting.
- Packaging units can use preset counts for batch handling.
- Warehouse operators can benefit from controlled material movement.
- Maintenance personnel can monitor I/O states for troubleshooting.
- Students gain practical experience in ladder logic and sequential control.

1.5 Objectives and Scope of the Project

Objectives:

1. Design a PLC-based conveyor-control system using Mitsubishi FX5U.
2. Develop a Ladder Diagram in GX Works3.
3. Implement start/stop and run-state memory.
4. Define object-detection or simulated detection input.
5. Count valid pulses and provide reset functionality.
6. Indicate completion at the preset count.
7. Prepare a design that can later be connected to a physical conveyor and sensor.

Scope:

The report covers control design, proposed I/O allocation, ladder structure, counting, reset, and completion indication. Pushbuttons and lamps may demonstrate the logic if supported by the actual training-kit wiring. Physical conveyor movement and automatic detection are not claimed until corresponding hardware is installed andtested.

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction

Conveyor automation combines a transport mechanism with electrical control, sensors, and a controller. PLCs are commonly used because sequence logic can be modified and monitored without redesigning the complete control circuit.

2.2 Literature Survey Table

Sr. No.	Title of Paper / System	Authors & Year	Technology Used	Limitations / Research Gap
1	*Size Based Counting of Moving Objects on Conveyor Belt using PLC*	Neeta Thune et al. (2025)	PLC, industrial vision camera, conveyor system and HMI	The system uses vision-based detection, which may require additional hardware and configuration compared with a basic digital-input counting system.
2	*PLC Counter System on Conveyor with Infrared Sensor as Input*	Roy Morales Gultom and Yuliarman Saragih (2024)	PLC, infrared sensor, GX Works2 and counter logic	Object counting depends on correct sensor installation and detection. The implementation uses a physical infrared sensor.
3	*PLC Program for Counting Moving Objects on Conveyor*	InstrumentationTools (2023)	PLC ladder logic, proximity sensor and up-counter	The example focuses on basic object counting. Additional system integration and hardware-specific configuration may be required for a complete conveyor application.
4	*Design and Development of a PLC-Based Automatic Object Sorting System*	RSIS International (2018)	PLC, ladder logic, conveyor, photoelectric and proximity sensors	The system requires multiple sensors and sorting hardware. A simpler demonstration can focus on start-stop control and object counting before adding sorting functions.

2.3 Problem Definition

Manual conveyor operation and counting require attention and may result in missed or duplicated counts. The proposed solution combines run control, object counting, reset, and count-completion indication in a PLC ladder program. Physical performance depends on installing and testing a compatible sensor, conveyor motor, and electrical interfaces.

2.4 Feasibility Study

- Technical feasibility: FX5U and GX Works3 can support a basic digital control sequence; actual CPU configuration, addressing, instruction syntax, and wiring must be checked.
- Operational feasibility: the sequence is straightforward and can be demonstrated with input pushbuttons and output indicators where the training setup permits.
- Economic feasibility: the available training kit and computer can support a basic demonstration. A physical conveyor requires additional sensor, motor, interface, and protection hardware.
- Educational feasibility: the project supports learning of addressing, ladder logic, counters, monitoring, and sequential control.

2.5 Methodology Used

The system follows a 8-stage operational methodology:

1. Requirement analysis and definition of control functions.
2. I/O address planning.
3. Ladder development in GX Works3.
4. Program syntax and address checks.
5. Input polarity and wiring verification.
6. Controlled pulse-counting demonstration.
7. Count completion and reset testing.
8. Documentation of actual observations.

CHAPTER 3: ANALYSIS AND PLANNING

3.1 Introduction

The system is divided into input handling, run control, object counting, completion indication, and counter reset. Planning the functions and device addresses before programming makes the logic easier to test and maintain.

3.2 Project Planning

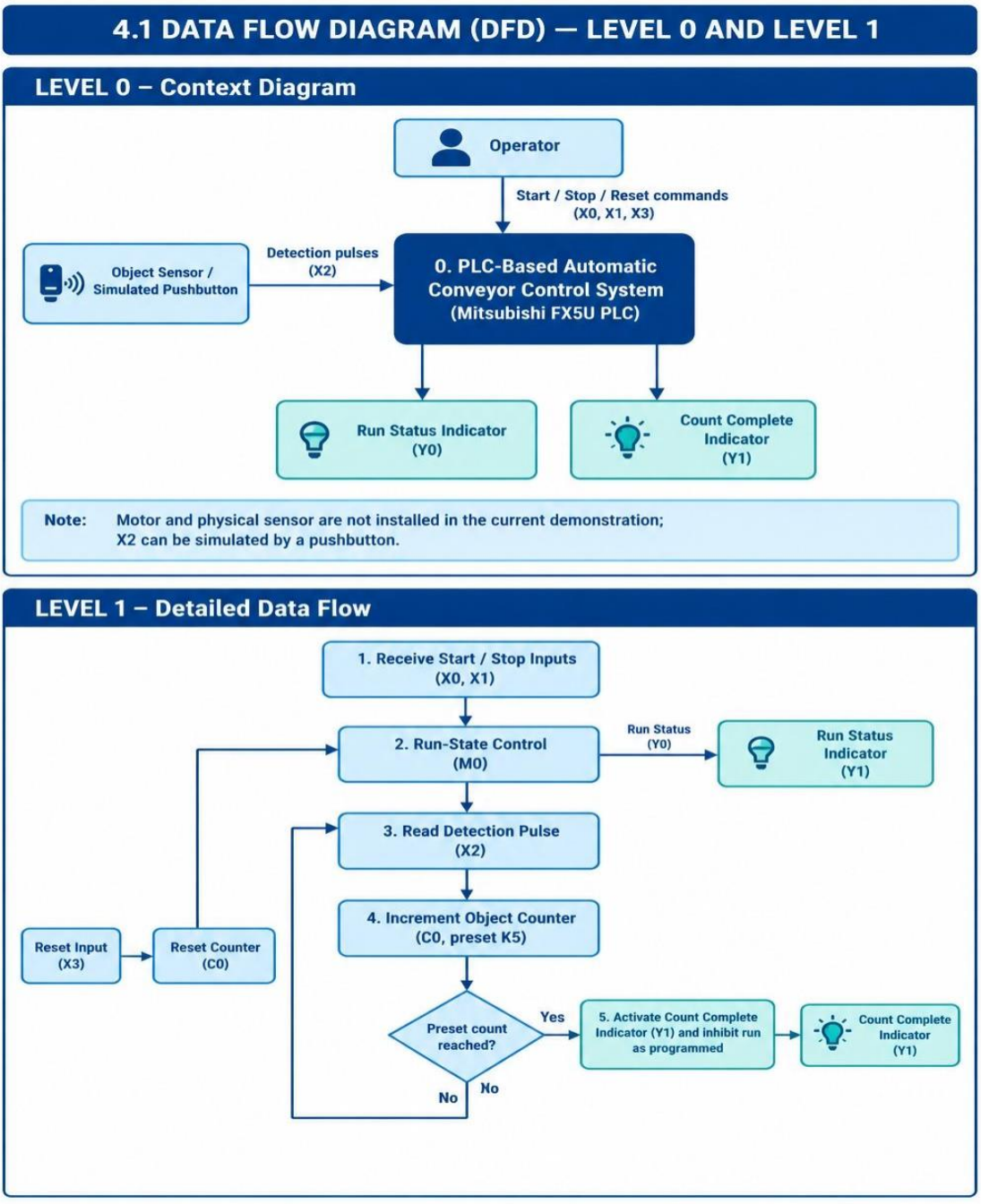
Table 2: Project Planning & Phase Breakdown

Phase	Activity Description	Deliverable	Duration
Phase 1	Requirement Analysis & Literature Review	Problem Statement and System Requirements	Weeks 1–2
Phase 2	PLC I/O Mapping and System Design	I/O Allocation Table, Block Diagram and Flowchart	Weeks 3–4
Phase 3	Ladder Logic Development in GX Works3	PLC Ladder Program	Weeks 5–7
Phase 4	Counter Logic and Reset Implementation	Object Counting and Reset Sequence	Weeks 8–9
Phase 5	Program Verification and Demonstration	Verified Logic or Simulation Results, subject to testing	Weeks 10–11
Phase 6	Documentation and Report Preparation	Final Prototype & Capstone Report	Week 12

CHAPTER 4: DESIGN & IMPLEMENTATION

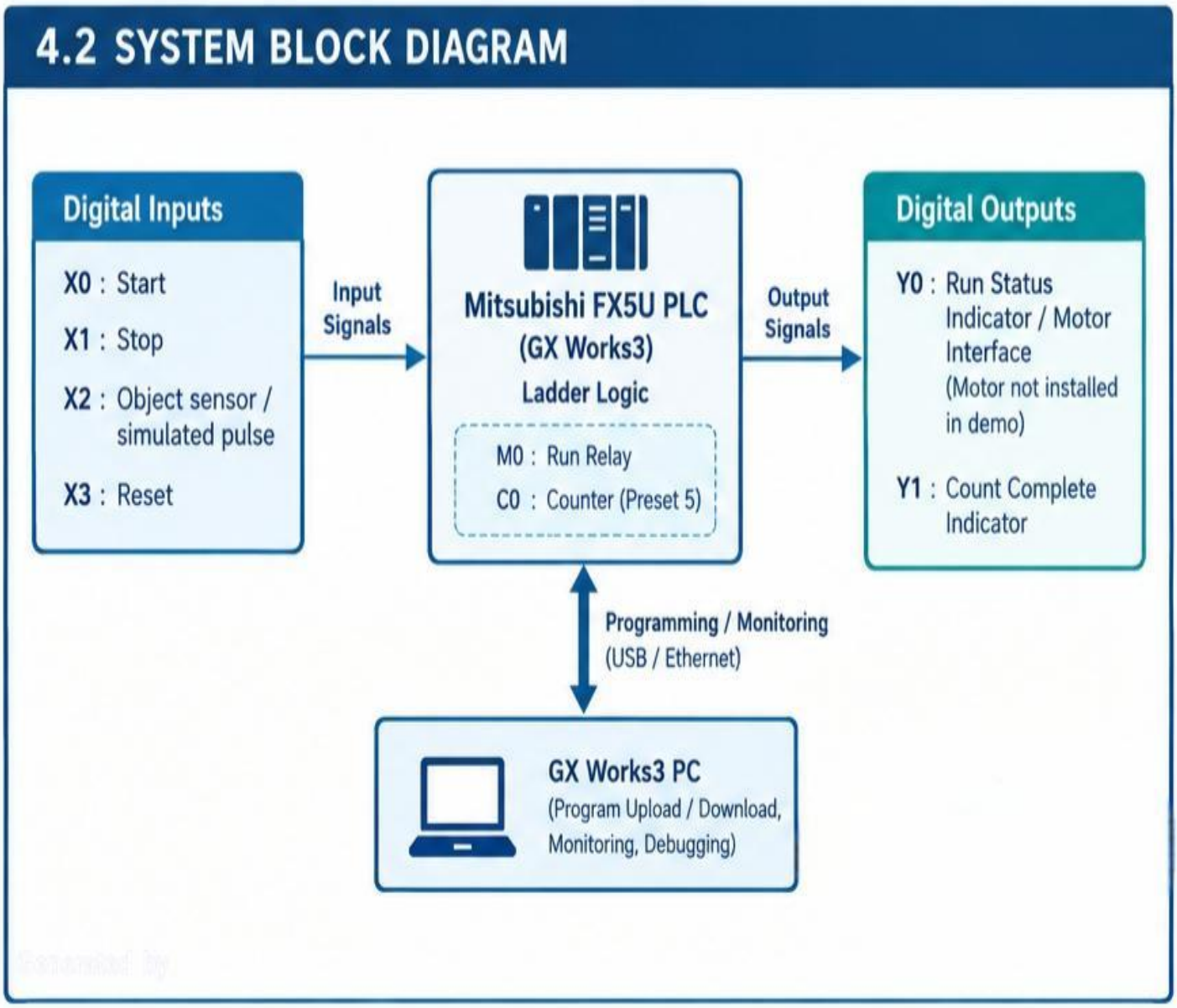
4.1 Data Flow Diagram (DFD)

Figure 1: DFD Level 0 and Level 1



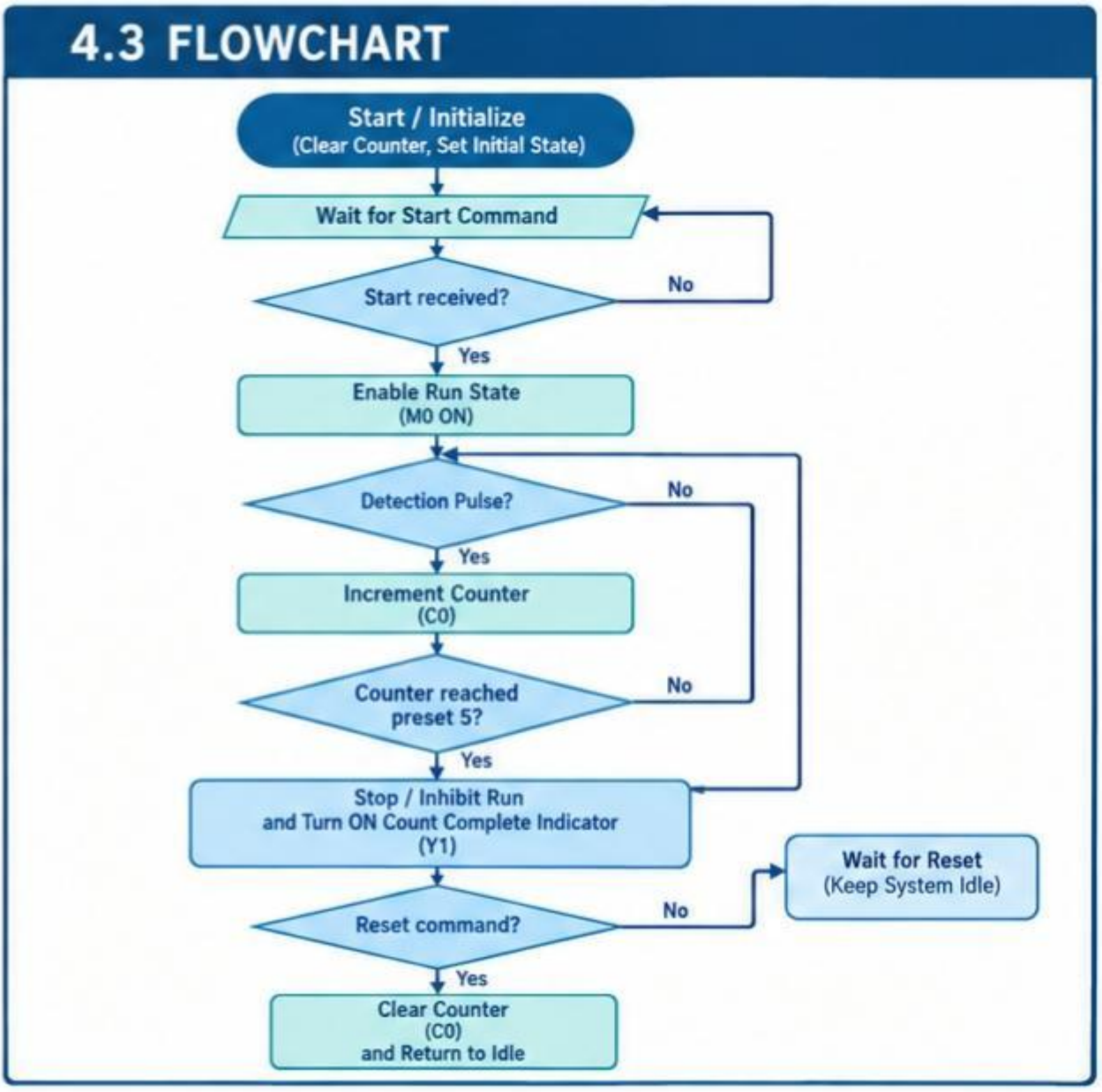
4.2 Block Diagram

Figure 2: System Block Diagram



4.3 Flow Chart

Figure 3: Sequence Flow Chart

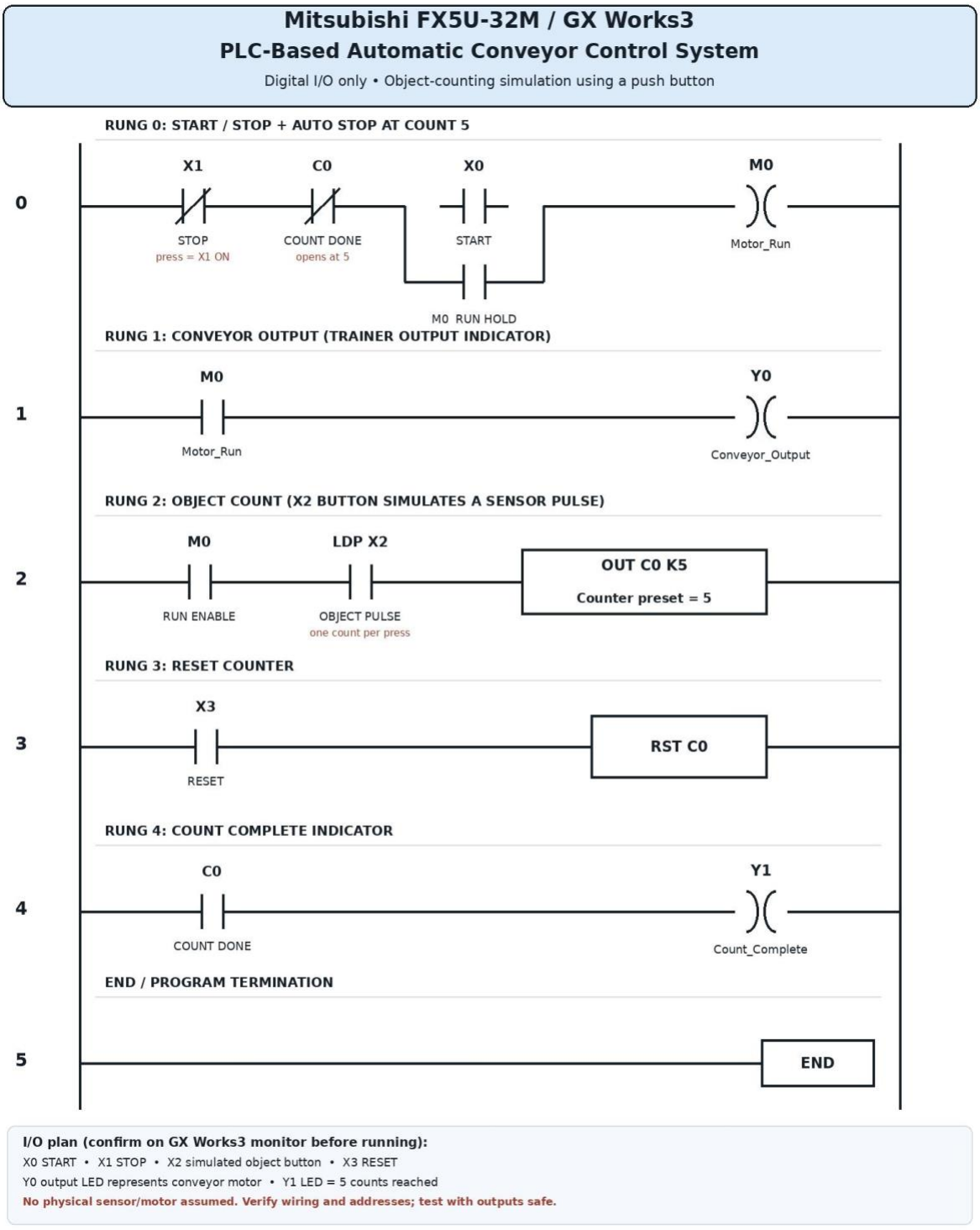


4.4 UML & PLC Ladder Diagram

Table 3: Mitsubishi FX5U PLC I/O & Internal Relay Allocation

Device Tag	Type	Function / Description
X0	Digital Input	Conveyor Start Pushbutton
X1	Digital Input	Conveyor Stop Command
X2	Digital Input	Object Detection Input (simulated using a pushbutton)
X3	Digital Input	Counter Reset Pushbutton
M0	Internal Relay	Conveyor Run Command / Master Run Latch
C0	Counter	Counts object-detection input pulses
Y0	Digital Output	Conveyor Run Status Indicator (motor output in future hardware implementation)
Y1	Digital Output	Count-Complete Indicator
END	Program Instructions	End of PLC program

Figure 4: Mitsubishi FX5U PLC Ladder Diagram (MELSOFT GX Works3 Forma



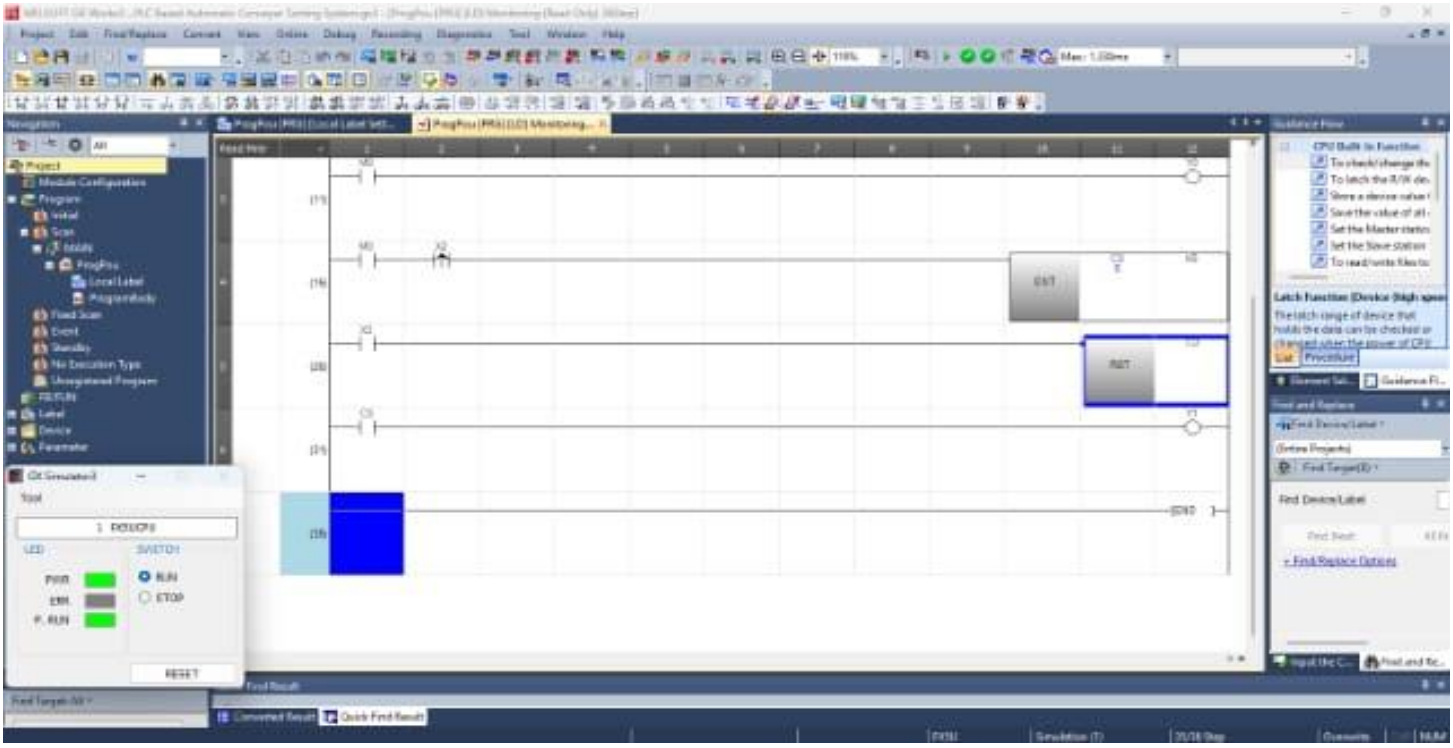
CHAPTER 5: RESULTS & DISCUSSION

5.1 Actual Results

The project is designed to demonstrate conveyor run control and object counting using Mitsubishi FX5U PLC ladder logic. The intended operating sequence includes start-stop control, simulated object-detection pulses, counter operation, count-completion indication and counter reset. The following behaviours are to be checked during program verification:

- Start/Stop Control: Verify that the run-status output responds correctly to the start and stop commands.
- Object Counting: Apply simulated detection pulses through X2 and check whether the counter increments correctly.
- Count Completion**:** Verify that Y1 activates when the configured counter preset is reached.
- Counter Reset: Operate X3 and check whether the counter returns to its initial value.
- Output Indication: Confirm that Y0 represents the run command and Y1 represents count completion.

The available setup does not include a physical conveyor motor or object-detection sensor. Therefore, the current demonstration is limited to the PLC training kit inputs and output indicators. Physical conveyor operation and sensor-based detection remain part of the future implementation.



5.2 Future Scope

1. Physical conveyor and motor-control integration.
2. Automatic detection using a compatible sensor.
3. Configurable batch sizes.
4. Automatic sorting using additional sensors and actuators.
5. HMI display of count, run status, and completion.
6. Production data logging.
7. Fault detection with appropriate feedback and protection devices.
8. Emergency-stop hardware, guards, interlocks, and motor protection.

CHAPTER 6:CONCLUSION

The PLC-Based Automatic Conveyor Control System provides a structured introduction to conveyor automation using the Mitsubishi FX5U PLC and MELSOFT GX Works3. The proposed design combines start/stop control, internal relay logic, object counting, counter reset, and count-completion indication. The project highlights the importance of edge detection, correct device addressing, compatible counter instructions, and verification of physical wiring. The available training setup may demonstrate the logic using pushbuttons as simulated detection signals and lamps as output indicators, subject to its actual wiring. Real object detection and conveyor movement require compatible sensor and motor hardware and practical testing.

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